A BOUNDED BELOW PSEUDO-COHERENT COMPLEX WHOSE FIBRE VANISHES IN LOW DEGREES BECOMES PERFECT AFTER LOCALIZING

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Allowed standard prerequisites (editorial).

- After replacing the ring by a localization at a suitable element, a pseudo-coherent complex splits as the direct sum of its truncation below a degree and its truncation above that degree, the latter being perfect. invoked at proof lines 20020-20024. Produces the splitting of the complex and the perfectness of the upper truncation.
- A perfect complex whose fibre over a prime is a finite free complex concentrated in degrees where the dimensions are prescribed can be represented over a suitable localization by the complex with those free modules. invoked at proof lines 20027-20029. Turns the perfect object into the displayed bounded complex of free modules with the prescribed ranks.

Source reference. Original lemma 068U: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19867–19867. Invoked at proof lines 20020-20024 to split the complex and to get the perfectness of the upper truncation; the vanishing of the lower truncation and the conclusion are inside the counted span.

Source reference. Original lemma 0BCD: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19594–19594. Used at proof lines 20027-20029 to produce the displayed free complex representing the object; the reduction to the perfect case that makes it applicable is inside the counted span.

Lemma 0.2. *Let R be a ring and let $\mathfrak{p} \subset R$ be a prime. Let K be pseudo-coherent and bounded below. Set $d_i = \dim_{\kappa(\mathfrak{p})} H^i(K \otimes^{\mathbf{L}}_R \kappa(\mathfrak{p}))$. If there exists an $a \in \mathbf{Z}$ such that $d_i = 0$ for $i < a$, then there exists an $f \in R$, $f \notin \mathfrak{p}$ and a complex*

$$\ldots \to 0 \to R_f^{\oplus d_a} \to R_f^{\oplus d_{a+1}} \to \ldots \to R_f^{\oplus d_{b-1}} \to R_f^{\oplus d_b} \to 0 \to \ldots$$

representing $K \otimes^{\mathbf{L}}_R R_f$ in $D(R_f)$. In particular $K \otimes^{\mathbf{L}}_R R_f$ is perfect.

Proof. After decreasing a we may assume that also $H^i(K^\bullet) = 0$ for $i < a$. By Lemma 068U after replacing R by R_f for some $f \in R$, $f \notin \mathfrak{p}$ we can write $K^\bullet = \tau_{\leq a-1}K^\bullet \oplus \tau_{\geq a}K^\bullet$ in $D(R)$ with $\tau_{\geq a}K^\bullet$ perfect. Since $H^i(K^\bullet) = 0$ for $i < a$ we see that $\tau_{\leq a-1}K^\bullet = 0$ in $D(R)$. Hence $K^\bullet$ is perfect. Then we can conclude using Lemma 0BCD. □

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